

Guía de Estudio: Applied AI & System Design

Esta guía contiene preguntas clave y conceptos fundamentales para el diseño de sistemas y la aplicación de inteligencia artificial, enfocada en la optimización de flujos de trabajo, bases de datos y arquitecturas RAG.

Applied AI 6

Pregunta: A RAG pipeline chunks customer support transcripts using a sliding window of 200 tokens with 50-token overlap. An evaluation shows retrieval recall is low for queries about issue resolutions, which are typically summarised in the final two sentences of each transcript. What is the most likely cause?

- The 200-token window is too small to capture the full resolution context, so the model generates incomplete answers even when the resolution chunk is retrieved.
- **The resolution sentences fall at the end of transcripts and may be split across the final two chunks, with neither chunk containing the full resolution statement needed to match the query (Respuesta correcta)**
- Sliding-window chunking distributes the transcript content evenly across chunks, making it impossible for any single chunk to score high on resolution-focused queries.
- The 50-token overlap is too large, causing the resolution sentences to appear in too many chunks and diluting their retrieval signal through repeated embedding.

Applied AI 5

Pregunta: A team applies a recursive character splitter to chunk a corpus of API reference documentation. Each page contains a function signature, a parameter table, and usage examples. Retrieval evaluation shows that queries for specific parameters frequently return chunks containing only the usage examples, missing the parameter table. Which chunking change most directly addresses this?

- Reducing the maximum chunk size so that parameter tables are never combined with usage examples in a single chunk, keeping each conceptual section separate.
- Applying a semantic chunking model that automatically identifies topic boundaries in the text and splits at points of maximal semantic shift.
- **Splitting each page by its structural sections – signature, parameter table, examples – so each chunk maps to a self-contained documentation element (Respuesta correcta)**
- Increasing the overlap between chunks so that the parameter table content appears in the preceding and following chunks, giving retrieval multiple chances to surface it.

System Design 48

Pregunta: A data engineering team must choose a database for a new application that processes 10,000 online retail orders per second, each requiring an immediate stock check and balance update. Which system type is most appropriate, and why?

- A document store in OLAP mode, because each order record maps naturally to a JSON document and OLAP document stores support high-throughput writes with flexible schema validation at scale.
- A data warehouse with micro-batch loading, because data warehouses are optimised for high-write throughput and can ingest 10,000 records per second using streaming insert APIs.
- An OLAP columnar store, because its columnar compression reduces the storage footprint for high-volume order data and its vectorised execution engine handles the concurrent writes efficiently.
- **An OLTP relational database, because the workload requires low-latency point reads, concurrent single-record updates, and ACID transactions to maintain stock and balance consistency (Respuesta correcta)**

System Design 42

Pregunta: A data engineering team loads 10 TB of daily data into Snowflake and runs 200 concurrent analytical queries throughout the business day. Peak query times are 9-11 AM and 2-4 PM. Which Snowflake feature most directly addresses the peak concurrency load without over-provisioning for off-peak hours?

- Configuring a single auto-suspend virtual warehouse that scales down to zero during off-peak hours, eliminating idle compute cost between the two peak windows.
- Increasing the virtual warehouse size from Medium to 4X-Large permanently, providing enough compute capacity to handle 200 concurrent queries at peak without any dynamic scaling.
- Enabling Snowflake's query result cache for all users so repeated queries return cached results instantly, reducing the number of queries that require virtual warehouse compute resources.
- **Creating separate virtual warehouses for loading and querying, and enabling multi-cluster warehouses on the query warehouse so it auto-scales during peak concurrency windows (Respuesta correcta)**

System Design 39

Pregunta: A team runs a PostgreSQL database with a table that receives 50,000 INSERT operations per second from an event stream. They add 5 additional indexes to support new analytical queries. Write throughput drops by 60%. What is the most likely cause?

- The 5 new indexes created a circular foreign key dependency that PostgreSQL must validate on every INSERT, increasing lock contention across the table's constraint graph.
- PostgreSQL's write-ahead log is filling faster than it can be flushed to disk, causing write stalls while the WAL buffer waits for I/O to complete before new writes can be accepted.
- **Each INSERT must now update 5 additional index structures in addition to the heap, multiplying the I/O cost per write and reducing overall write throughput proportionally (Respuesta correcta)**
- PostgreSQL's auto-vacuum process is running more frequently because the additional indexes generate more dead tuple records per INSERT, competing with write operations for I/O bandwidth.

System Design 38

Pregunta: What is the primary guarantee provided by ACID transactions in a relational database?

- ACID transactions guarantee that all queries on a database table complete within a configurable time limit, preventing long-running queries from blocking other concurrent database operations.
- ACID transactions guarantee that the database automatically partitions large tables into shards when row counts exceed a threshold, maintaining query performance as data volume grows.
- ACID transactions guarantee that a relational database can store any data type natively without schema definition, allowing flexible document-like records alongside structured tabular data.
- **ACID transactions guarantee that a set of database operations either all succeed or all fail atomically, leaving the database in a consistent state even in the event of errors or crashes (Respuesta correcta)**

System Design 34

Pregunta: A RAG pipeline's retrieval step returns highly relevant chunks, but the final LLM responses are still inaccurate for numerical data (dates, percentages, amounts). Which change most directly addresses this specific failure mode?

- Increasing the number of retrieved chunks from top-5 to top-20 so the model has more numerical data to cross-reference and can select the most consistent values across all retrieved chunks.
- **Passing the retrieved chunks to the LLM with explicit instructions to quote numerical values directly from the source text, and including metadata like the source document and page number (Respuesta correcta)**
- Switching the similarity metric from cosine similarity to dot product distance, since dot product rewards vectors with higher magnitude which correlates with denser factual content in the chunks.
- Adding a post-processing step that validates each numerical value in the LLM's response against a predefined list of acceptable ranges, flagging responses that contain out-of-range numbers.

System Design 25

Pregunta: A data engineering team uses an LLM to extract structured fields from free-text incident reports. Outputs must be consistent and reproducible — the same report processed twice must always produce identical JSON. Which combination of LLM parameter settings best achieves this?

- Set temperature to 0 and top_p to 0.5 to enable partial randomness only within the top half of the probability mass, producing outputs that are mostly deterministic with limited variance.
- Set temperature to 1.5 and top_p to 0.9 to generate high-entropy outputs that explore more of the probability space, increasing the chance of finding the globally optimal extraction.
- Set temperature to 0.7 and top_k to 50 to balance determinism with diversity, allowing the model to occasionally surface alternative field values from ambiguous report text.
- **Set temperature to 0 to make token selection deterministic, ensuring the same input always produces the same output with no sampling-based variance (Respuesta correcta)**

System Design 5

Pregunta: A REST API endpoint returns HTTP 200 OK with a body containing {"error": "user not found"} when a requested user does not exist. A consuming service logs no errors despite the lookup failing. What is the root cause?

- The consuming service is rate-limited and receives cached 200 responses from a CDN that does not inspect the body content before serving cached results to the client.
- The endpoint lacks authentication middleware, so unauthenticated requests return 200 with synthetic user data rather than the real error body when a user is not found.
- **The consuming service checks only the HTTP status code for error detection; returning 200 for a missing resource violates REST conventions and suppresses error handling in the client (Respuesta correcta)**
- The consuming service's HTTP client library strips the response body when the status code is 200, preventing the error message from being accessible in the application code.

Data Engineering 39

Pregunta: A data engineering team designs a test strategy for a pipeline that ingests, transforms, and loads customer orders. They have unit tests for individual transformation functions but no integration or contract tests. A schema change in the source system causes silent data corruption in the output table that goes undetected for two weeks. Which testing addition most directly prevents this class of failure?

Opciones:

- Adding more unit tests for each transformation function to increase code coverage and catch edge cases in the transformation logic that could produce incorrect output values
- Adding end-to-end regression tests that compare the full pipeline output to a golden dataset produced before any schema changes, flagging any deviations from the expected output
- **Adding schema contract tests at the pipeline's ingestion boundary that assert expected column names, types, and non-null constraints, failing the run immediately when the source schema changes**
- Adding static type annotations to all transformation functions and running mypy as part of CI to catch type mismatches that could result from upstream schema changes

Data Engineering 31

Pregunta: A data pipeline ingests customer records containing PII and writes them to a data lake. Multiple downstream teams query the data lake for analytics. Which access control approach most consistently enforces PII schema restrictions across all consumers?

Opciones:

- Distributing PII-redacted copies of the data to each team and instructing teams not to share their copies with other teams, relying on organisational policy to prevent unauthorised PII access
- Applying application-level filtering in each downstream pipeline's code so each team's queries explicitly exclude PII columns, relying on code review to catch violations
- **Enforcing column-level or row-level access control at the data lake's query layer so PII columns are masked or redacted automatically based on the querying user's role and permissions**
- Creating a separate PII-free dataset by running a batch anonymisation job nightly and directing all analytics teams to query the anonymised copy rather than the raw lake

Data Engineering 30

Pregunta: A data engineering team must share a dataset of patient records with a research partner for epidemiological analysis. The partner needs age, diagnosis code, and zip code. Direct identifiers (name, NHS number) must not be shared. Under GDPR, which data handling approach best satisfies both the research purpose and the minimum data principles?

Opciones:

- Share the full dataset with all columns encrypted at rest and instruct the research partner to retain the decryption key on their infrastructure under a data sharing agreement
- **Share only the required fields - generalised age band, diagnosis code, and region rather than zip code - with direct identifiers removed, applying data minimisation and k-anonymity principles**
- Share the full dataset with name and NHS number replaced by consistent hash values, allowing the partner to re-identify patients by cross-referencing their own patient database
- Share only name and NHS number in pseudonymised form and encrypt diagnosis codes, since diagnosis codes are more sensitive than demographic data and require stronger protection

Data Engineering 27

Pregunta: A pipeline that aggregates daily sales reports passes all data quality checks but stakeholders report that the sales totals seem too low. Investigation reveals that the pipeline only processes events received before 11:59 PM and misses orders submitted in the final 2 minutes of each day due to a processing lag. Which data quality dimension does this gap violate, and what is the appropriate fix?

Opciones:

- Accuracy, because the totals do not reflect the true value of orders placed on that day; fix by retraining the model that generates sales estimates to account for the 2-minute lag
- Consistency, because the pipeline's daily totals disagree with the source system's order management totals by the missing 2 minutes; fix by aligning the pipeline's ingestion window to the source system's day boundary
- **Completeness, because a systematic subset of daily orders is excluded from every aggregate; fix by extending the ingestion window or adding a reconciliation pass that captures late-closing orders**
- Timeliness, because the pipeline is not fast enough to process all orders within the same calendar day they are placed; fix by increasing the pipeline's processing speed to eliminate the 2-minute lag

Data Engineering 19

Pregunta: A company processes clickstream events using a nightly batch job. They want to migrate to stream processing to reduce latency but are concerned about handling out-of-order events. In their current batch job, late-arriving events are captured because the batch processes all events for the day regardless of arrival

order. Which stream processing mechanism provides equivalent late-event handling?

Opciones:

- Configuring the stream processor to buffer all incoming events in a priority queue sorted by event time before processing, ensuring events are always processed in exact event-time order
- Running a reconciliation batch job alongside the stream processor that corrects any aggregations affected by late events on a daily basis, combining both processing approaches
- **Configuring event-time watermarks and window functions that hold state open for a defined late-arrival tolerance, allowing late events to be included in the correct time window**
- Increasing the stream processor's micro-batch interval to match the daily batch window, making the stream processor functionally equivalent to the existing batch job

Data Engineering 8

Pregunta: In dimensional modelling, what is the defining characteristic of a fact table?

Opciones:

- A fact table stores descriptive attributes about business entities such as customers, products, or locations, structured to support efficient filtering and grouping operations
- A fact table stores a single row per business entity updated in place whenever an attribute changes, providing a current-state view of every entity in the domain
- A fact table stores the relationship between an agent and a policy in an insurance schema, linking the two entities through a compound primary key
- **A fact table stores measurements or events at a defined grain, with foreign keys to dimension tables and numeric measures that are aggregated in analytical queries**

Applied AI 31

Pregunta: A team benchmarks an LLM for a classification task by measuring average accuracy and average cost per request. They select the model with the better accuracy-to-cost ratio. After deployment, the cost-per-day significantly exceeds the benchmark projection. What is the most likely reason?

Opciones:

- The provider applied a pricing tier change between the benchmark date and deployment, increasing the cost per token for the selected model by a fixed percentage
- The model's output token count was not included in the cost calculation, and the deployed classifier produces longer explanations than the benchmark prompts expected
- **The benchmark used a subset of inputs that had shorter token lengths than the production distribution, underestimating average input token cost per request**
- The model's accuracy degraded after deployment due to input distribution shift, causing it to produce longer chain-of-thought traces that increased token consumption per request

Applied AI 22

Pregunta: What is the primary purpose of a human-in-the-loop checkpoint in an agentic pipeline that performs automated data edits?

Opciones:

- To reduce total pipeline latency by parallelising the human review step with other agent tasks that are not on the critical path of the reviewed action
- **To require explicit human approval before an irreversible or high-risk action is executed, preventing automated errors from propagating**
- To cache each approved output in a reusable store so that future similar operations can bypass the checkpoint and be applied without re-engaging a human reviewer
- To act as a fallback model that takes over from the primary LLM whenever it produces a low-confidence output requiring secondary generation before the pipeline continues

Applied AI 11

Pregunta: A data engineering team receives a mixed feed containing CSV sales records and JSON customer events. They must load both into a unified analytics table for reporting. Which data preparation step is most critical before loading?

Opciones:

- Validating that both file types use the same primary key format so that join operations between the sales and customer tables work correctly after loading
- Deduplicating records within each file type before loading so that any duplicate CSV row or repeated JSON event does not inflate the analytics table counts

- **Normalising field names, data types, and date formats across both sources so records are comparable and joinable in the unified table**
- Compressing both file types to a common format such as gzip to standardise storage encoding and reduce transfer time during the load process

Applied AI 8

Pregunta: A data team extracts structured fields from scanned invoice images using an OCR pipeline followed by an LLM extraction step. Invoice totals are frequently wrong by a single digit — e.g., \$1,234 extracted as \$1,284. Which pipeline stage is the most likely source of this error?

Opciones:

- The image preprocessing step is over-compressing the invoice image before OCR, reducing resolution to the point where multi-digit numbers are partially lost before any text is read
- **The OCR engine is misreading individual digits in low-contrast or degraded print regions, producing a single-character substitution that the LLM receives as correct input and faithfully extracts**
- The LLM extraction step is rounding numeric values to nearby round numbers based on statistical patterns in its training data, substituting plausible amounts for exact figures
- The LLM's context window is being exceeded by long invoices, causing the model to hallucinate the total field from values it observed earlier in the truncated context

Applied AI 7

Pregunta: A team chunks a corpus of research papers using paragraph boundaries and a 512-token limit. Papers regularly contain multi-paragraph mathematical derivations where each paragraph depends on the one before. Users report that answers to derivation queries are missing intermediate steps. Which chunking modification most directly preserves derivation continuity?

Opciones:

- Reducing the chunk size limit from 512 to 256 tokens to prevent long derivation paragraphs from being cut in the middle of an equation by the token limit
- Applying a topic modelling algorithm to each paper to identify derivation sections and attach their identity as a shared metadata tag on all their constituent chunks

- Switching from paragraph boundaries to sentence boundaries to produce finer-grained chunks that are less likely to contain more than one derivation step per chunk
- **Increasing overlap so each chunk carries the preceding derivation paragraph forward, preserving the reasoning chain across boundaries**

Applied AI 6

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